

SAKETH TALASU

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B.Tech Computer Science (AI & ML) undergraduate with a strong foundation in Machine Learning, Data Analytics, and Full-Stack Development. Proficient in building end-to-end solutions, from Computer Vision models to interactive data dashboards. Currently expanding expertise into Agentic AI and autonomous workflows to build intelligent, goal-oriented systems. Adept at leveraging Python, deep learning, and collaborative problem-solving to drive data-informed decision-making in fast-paced environments.

EDUCATION

B.Tech, CSE (AI & ML) , Anil Neerukonda Institute of Technology & Sciences — CGPA: 8.75/10	2027
Intermediate (MPC) , Tirumala Educational Institutions — 97%	2023
Class X (SSC) , Sri Chaitanya Junior College — 99%	2021

SKILLS

- **Programming Languages:** Python, C, JavaScript, SQL
- **AI/ML Frameworks:** TensorFlow, Scikit-Learn, XGBoost, OpenCV, RAG
- **Web & Cloud:** React, Node.js, FastAPI, MERN Stack, AWS, GCP, Azure
- **Databases:** SQL, MongoDB, SQLite
- **Core Competencies:** Machine Learning, Computer Vision, Explainable AI (XAI), AI Agents, Power BI

EXPERIENCE

AI & Data Engineering Intern <i>Covasant Technologies, Hyderabad</i>	May 2026 – June 2026
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- Developed an end-to-end **Predictive Maintenance** system for ACME Industries, integrating anomaly detection, failure prediction, and AI-driven diagnosis agents.
- Implemented **XGBoost** and **Isolation Forest** models with a multi-sensor feature engineering pipeline covering statistical, time-domain, and frequency-domain features.
- Built **RAG-based AI Agents** for failure diagnosis, fix recommendation, and automated ticket generation using **Databricks** for scalable data processing.

Research Intern Machine Learning <i>National Institute of Technology (NIT), Rourkela</i>	June 2025 – July 2025
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- Developed and optimized **Deep Learning** and **Computer Vision** models for Intelligent Transport Systems (ITS), implementing **CNN architectures** and advanced image preprocessing pipelines.
- Conducted performance evaluation using standard metrics to assess real-world feasibility in autonomous vehicle applications.
- Collaborated with research scholars on data-driven modeling for scientific and engineering problems.

PROJECTS

Predictive Maintenance Agent Stack — Python, XGBoost, Isolation Forest, RAG	Live Demo
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- Designed a feature engineering pipeline extracting statistical, time-domain, and frequency-domain features from multi-sensor data (vibration, temperature, current, pressure, acoustic).
- Built an **Isolation Forest** anomaly detection model and an **XGBoost** failure prediction model trained on simulated industrial sensor data.

- Developed AI agents for failure diagnosis, fix recommendation, and automated ticket generation using a **RAG**-based architecture over maintenance manuals and repair logs.

Traffic Sign Recognition System — MobileNetV2, FastAPI, React, TensorFlow

[Live Demo](#)

- Trained and fine-tuned a **MobileNetV2** model on the GTSRB dataset achieving **91%+** real-world accuracy, inspired by Advanced Driver-Assistance Systems (ADAS).
- Built a **FastAPI** REST API for real-time model inference with **SQLite** logging, and a responsive **React (Vite)** dashboard supporting drag-and-drop and live webcam inputs.
- Implemented **Explainable AI (Grad-CAM)** using OpenCV to generate dynamic heatmaps, providing visual transparency into the model's decision-making process.

Sudhari Nuts Sales & Analytics Platform — React, Node.js, SQL

[Live Demo](#)

- Architected a full-stack sales management system featuring role-based authentication and secure backend APIs.
- Engineered interactive dashboards with inventory tracking, profit estimation, and CSV/Excel export functionality.
- Integrated a sales forecasting module to provide data-driven business insights and inventory optimization.

CERTIFICATIONS

- Certified Cloud Practitioner – Amazon Web Services
- Power BI for Business Professionals – Infosys Springboard
- Python Certification – HackerRank
- Web Development Certification – Internshala

EXTRA-CURRICULAR ACTIVITIES / LEADERSHIP

- **Student Organizer**, Ignite Hack (INVENTUM 2K26) *2026*
 - Organized and executed an 8-hour inter-collegiate hackathon, coordinating event planning, logistics, and operations for 287 participants across 80 teams
 - Collaborated with faculty coordinators and industry partners to ensure smooth event delivery and active participant engagement
 - Awarded an appreciation certificate for leadership, coordination, and successful execution of the event
- **Team Participant**, National Entrepreneurship Challenge (E-Cell, IIT Bombay) *2025*
 - Participated in a 6-month nationwide entrepreneurship program focused on building and strengthening campus entrepreneurship ecosystems
 - Contributed to the NEC 2025 Basic Track by collaborating with team members on strategic initiatives and organizational development tasks
 - Gained exposure to entrepreneurial leadership, networking, and structured problem-solving